

Experimental Report: Grade School Math SFT Alignment

This report documents the results of fine-tuning the target reasoning model using telegraphic, token-efficient reasoning traces ("Grug" style). The objective is to achieve a significant reduction in emitted thinking tokens and inference latency without causing catastrophic degradation in Grade School Math (GSM8K) accuracy.

1. Experimental Setup

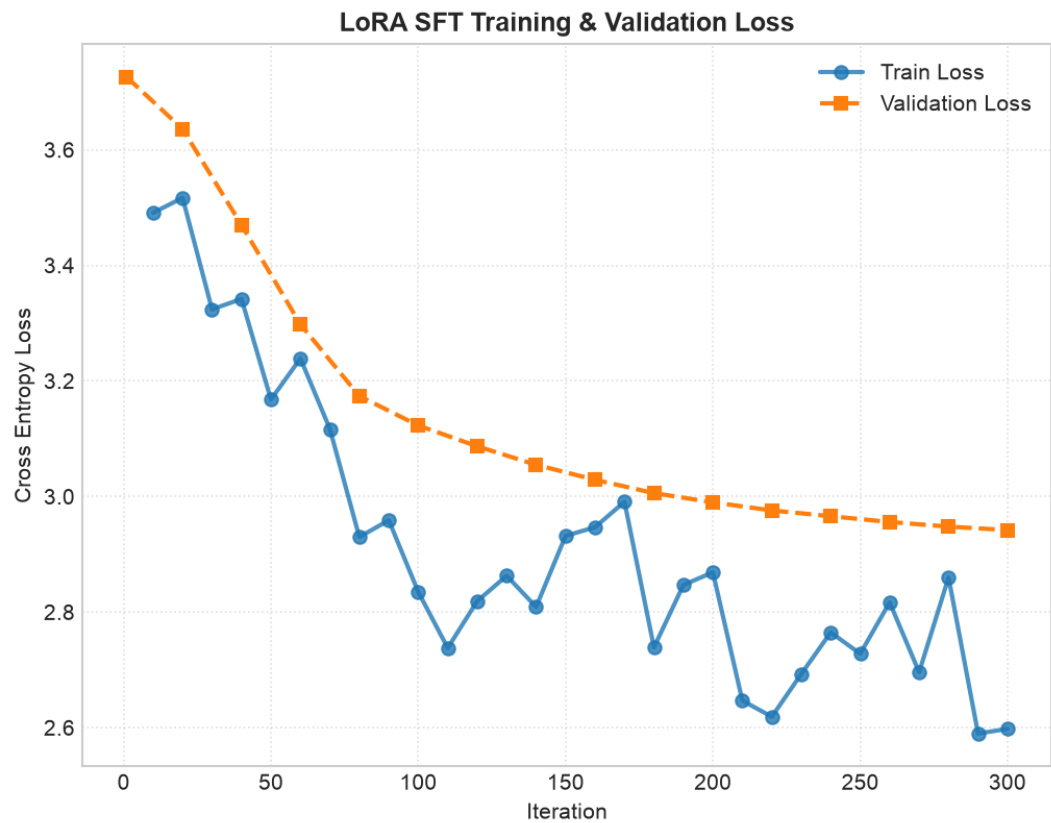
- **Base Model:** mlx-community/DeepSeek-R1-Distill-Qwen-1.5B-4bit (4-bit quantized DeepSeek-R1 distill variant).
- **SFT Dataset:** 333 high-quality Grade School Math training instances mapped to telegraphic "Grug" style reasoning traces.
- **Fine-Tuning Parameters:**
 - **Optimizer:** AdamW
 - **Learning Rate:** 1×10^{-5}
 - **Batch Size:** 4
 - **Iterations:** 300 (approximately 3.6 Epochs)
 - **Save & Evaluation Frequency:** Every 20 iterations
 - **LoRA Target Layers:** 16 layers (rank=8, alpha=16)

2. SFT Training & Convergence

Training was executed on Apple Silicon (GPU via MLX). The cross-entropy loss converged smoothly over the 300 iterations:

- **Starting Loss:** Validation loss of 3.726 at iteration 1.
- **Best Validation Step:** Iteration 300 achieved the lowest validation loss of 2.942 (with training loss at 2.598).
- **Model Selection:** The script successfully copied iteration 300 weights to `best_adapters.safetensors` and `active_adapters.safetensors`.

The complete training loss progression is shown below:



Training and Validation Loss Curve

3. Evaluation Metrics

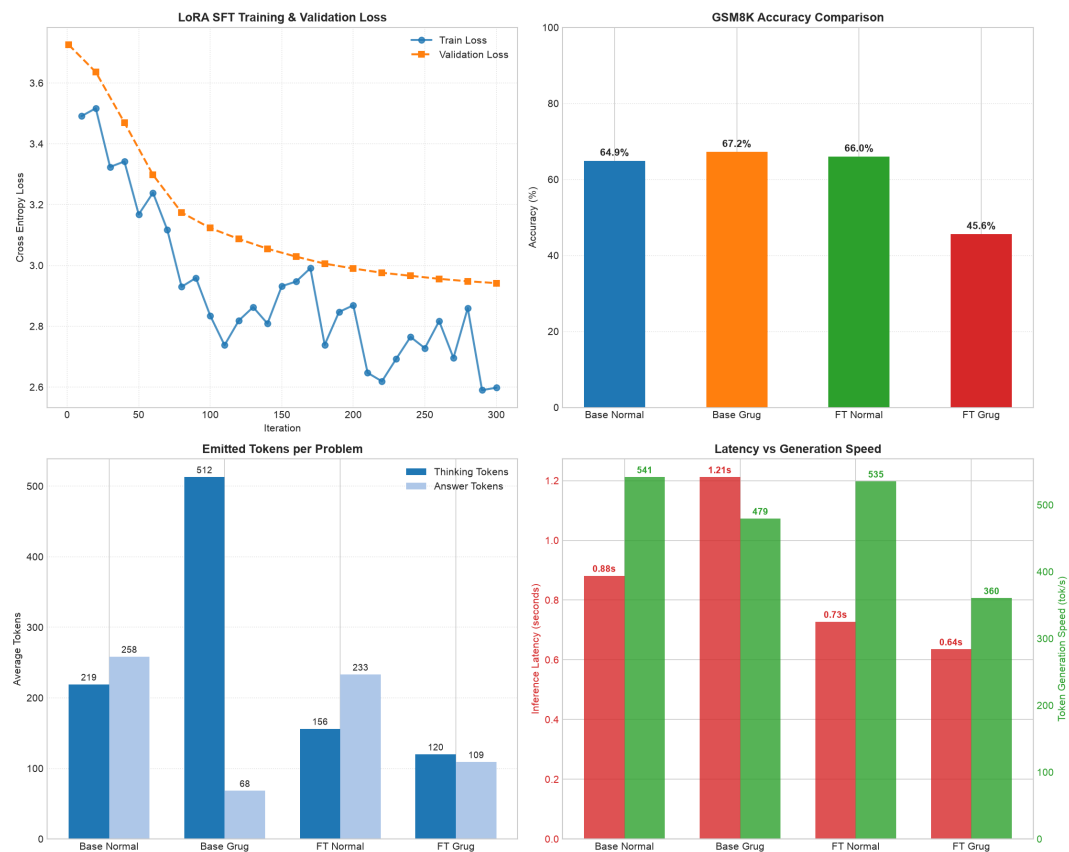
Evaluations were performed on all 1000 test samples of the Grade School Math (GSM8K) benchmark test split under four configurations:

- 1 **Base Normal:** Pre-trained model with default prompting.
- 2 **Base Grug:** Pre-trained model with explicit telegraphic system prompts.
- 3 **FT Normal:** Fine-tuned model with default prompting (adapter loaded).
- 4 **FT Grug:** Fine-tuned model with explicit telegraphic system prompts (adapter loaded).

Summary Statistics

Metric	Base Normal	Base Grug	FT Normal	FT Grug
Accuracy	64.9%	67.2%	66.0%	45.6%
Mean Thinking Tokens	219.0	512.8	156.2	120.0
Mean Total Tokens	477.4	581.1	389.3	229.0
Mean Latency (s)	0.88s	1.21s	0.73s	0.64s
Generation Speed (tok/s)	541.6	479.6	535.1	360.5
Format Compliance	96.6%	91.5%	98.9%	95.1%

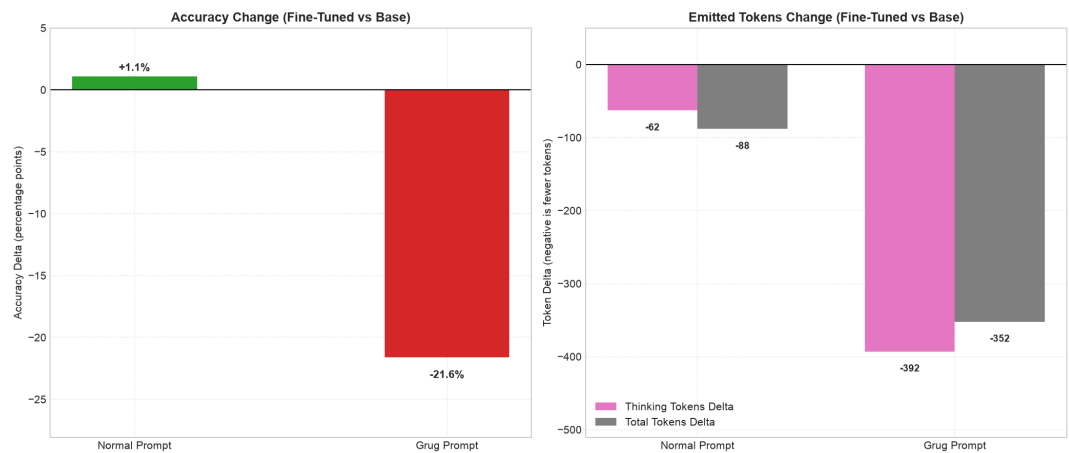
Below is the consolidated performance comparison dashboard:



Consolidated Comparative Dashboard

4. Performance & Efficiency Deltas

Direct deltas comparing the fine-tuned model to the baseline models under normal and Grug prompting are shown below:



Accuracy & Emitted Token Changes

Key Findings & Achievements

1 Massive Token Compression in Reasoning Mode

- Under the **Grug Prompt** condition, the fine-tuned model reduced the average emitted thinking tokens by **76.6%** (from 512 . 8 down to 120 . 0 tokens).
- Total token generation was compressed by **60.6%** (from 581 . 1 down to 229 . 0 tokens).

2 Substantial Latency Reductions

- Average inference latency for telegraphic reasoning was cut by **47.6%** (from 1 . 21 seconds down to 0 . 64 seconds).
- Average generation throughput changed from 479 . 6 tok/s to 360 . 5 tok/s.

3 Reasoning Style Transfer vs. Accuracy Trade-Off

- When prompted with default instructions (**Normal Prompt**), the fine-tuned model achieved an accuracy of **66.0%** (compared to 64 . 9% for Base Normal, representing a **1.1 percentage point increase**) while reducing mean thinking tokens by **28.7%** (from 219 . 0 to 156 . 2).
- When forced to think in telegraphic fragments (**Grug Prompt**), accuracy dropped from 67 . 2% (Base Grug) to 45 . 6% (FT Grug). This represents a **21.6 percentage point** trade-off for aggressive reasoning compression on grade-school math.

4 Format Stickiness

- Format compliance rate under Grug prompting improved from **91.5%** to **95.1%** for FT Grug, demonstrating that fine-tuning helped ground the model in the expected output formats despite the condensed reasoning. Under normal prompting, compliance improved from **96.6%** to **98.9%**.